Fiscal Year:	FY 2005	**Task Last Updated:**	FY 12/15/2004
PI Name:	Ronney, Paul D. Sc.D.		
Project Title:	Thermal Transportation Based Microscale Propulsion and Power Generation Devices		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Droplets, sprays, particles, dust clouds, nanoparticles		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Los Angeles	**State:**	CA
Zip Code:	90089-1453	**Congressional District:**	33
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	10/01/2003	**End Date:**	09/30/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA GRC
Contact Monitor:	Struk, Peter	**Contact Phone:**	216-433-5948
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	1		
COI Name (Institution):			
Grant/Contract No.:	NNC04GA31G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	In this work we introduce a self-sustaining, self-pressurizing catalytic combustion-driven thermal transpiration pump and reactor. Thermal transpiration occurs in capillary tubes or porous material when (1) a temperature gradient is maintained across the tube or pore and (2) the tube or pore diameter is comparable to the mean free path of gas molecules. Under such conditions a pressure gradient is produced in the same direction as the temperature gradient. Thermal transpiration (thus pressurization) is accomplished using membranes made of nanoporous materials (aerogels and glass-fiber filters) that have pore sizes comparable to the mean free path of gas molecules and have very low thermal conductivity, thereby enabling temperature gradients to be sustained with minimal thermal power. Catalytic combustion of hydrogen-air mixtures on the downstream side of the membrane provided the thermal power for self-sustaining pumping of the reactants. The objective of this work is to determine whether pumping and propulsion based on this catalytic combustion driven thermal transpiration concept is feasible and if so to determine the optimal design and operating conditions for devices exploiting this concept. Due to recent advances in MEMS technologies, there is considerable interest in		

	miniaturizing thermochemical systems for electrical power generation, gas chromatography, gas separation and chemical reforming, and propulsion for micro air vehicles. These systems all require gas pressurization and/or vacuum pumping systems that in turn require devices with moving parts that are difficult to miniaturize due to problems with friction losses, manufacturing tolerances, sealing, etc. The proposed device accomplishes pumping with no moving parts and using thermal rather than electrical energy as its feedstock.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed devices are applicable to portable electronic devices, microcombustors, microreactors and micro propulsion devices on earth as well as in space.
Task Progress:	Simple test fixtures were constructed and found to produce steady pumping for at least several hours of continuous operation when supplied with electrical heating. Measured flow rates and pressures as a function of the imposed temperature differential across the membrane are in reasonably good agreement with theoretical predictions by Vargo et al. (1999) for both types of membranes. As little as 1.1 Watt of fuel enthalpy would self-sustain the device. This proof-of-concept apparatus demonstrates the first-ever example of a gas pump with no moving parts that derives its energy for pumping from the gas itself with no external energy source.
Bibliography Type:	Description: (Last Updated: 01/29/2007)
Patents	Application 10/857,670 Pending May-2004 Ronney, P. D. "Thermal transpiration pump for gaseous material driven by chemical reaction"
Presentation	Ochoa, F., Ronney, P. D. "A thermal transpiration-based self-pressurizing mesoscale combustor" 4th Joint Meeting, U.S. Sections, Combustion Institute, Philadelphia, PA, March 21-23, 2005. Mar-2005